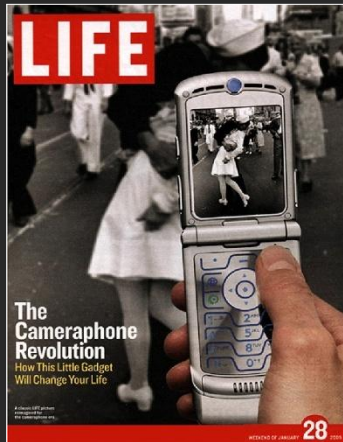


Computer Vision

Dr.Eng. Mahmoud Abu_Alfutuh

Computer Vision



Today

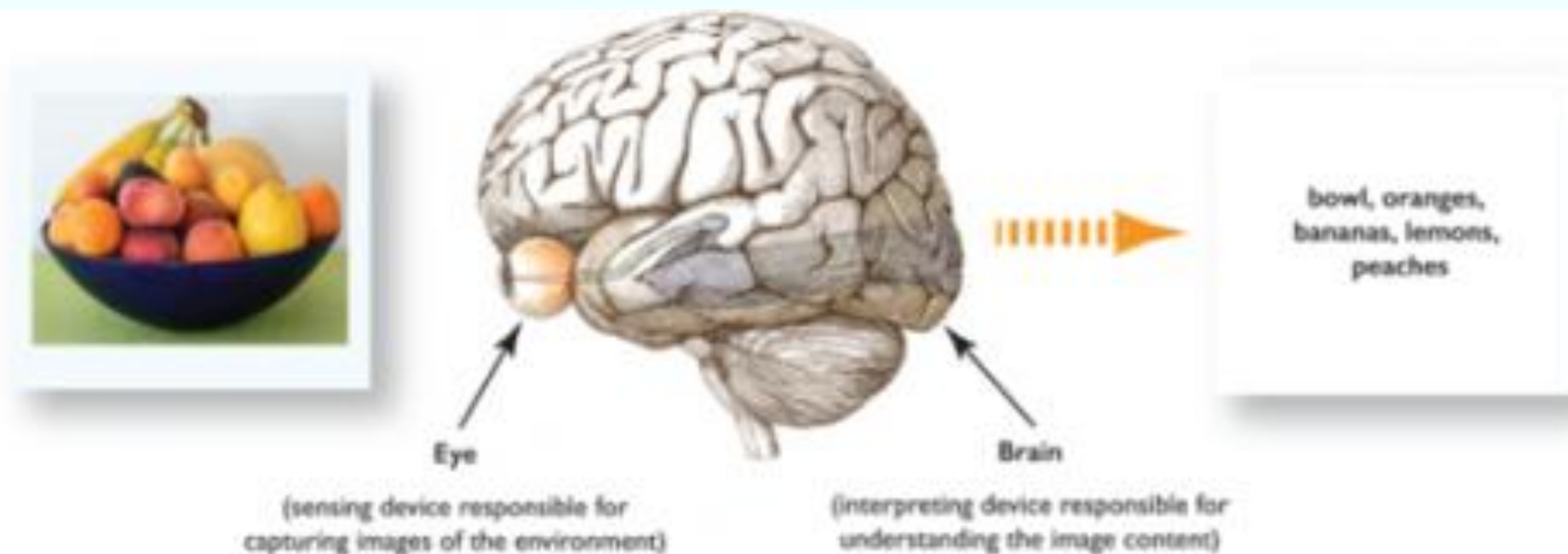
1-Introduction

- a) What is computer vision?
- b) How does computer vision work?
- c) The goal of computer vision
- d) Applications of Computer Vision
- e) Course overview

2-Machine Vision Systems

What is computer vision?

Computer vision is the field of computer science that focuses on creating digital systems that can process, analyze, and make sense of visual data (images or videos) in the same way that humans do.

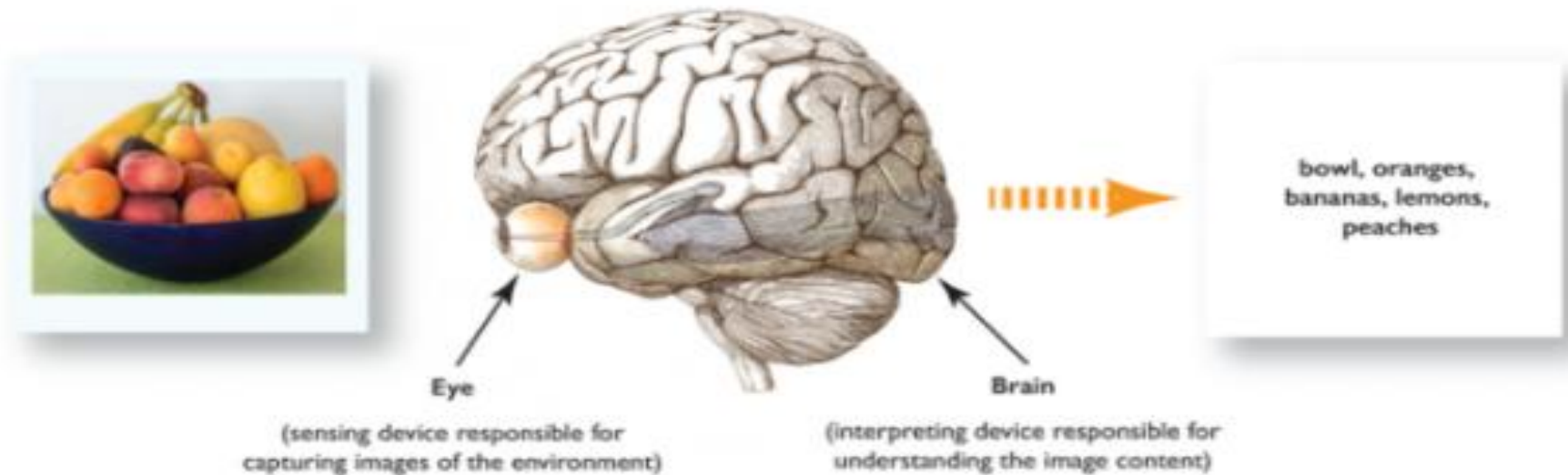


What is computer vision?

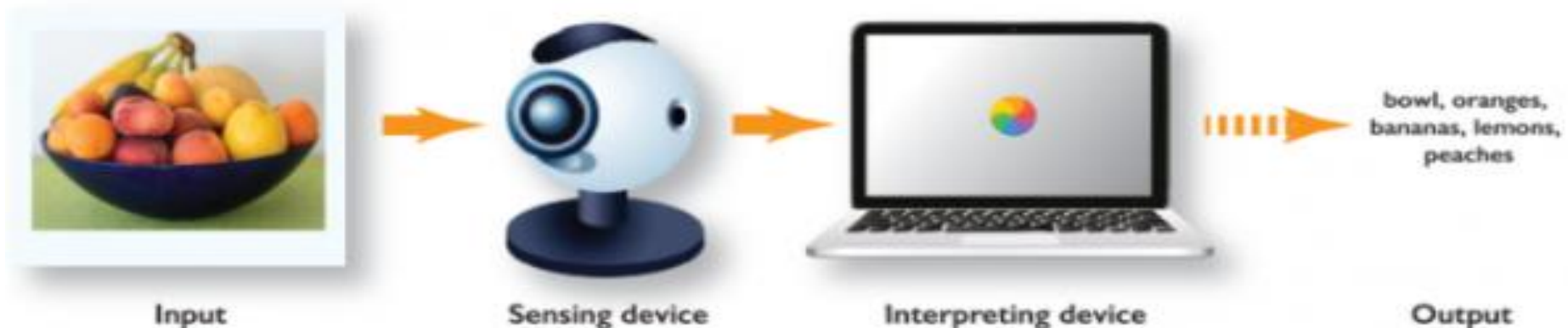
Technically, machines attempt to retrieve visual information, handle it, and interpret results through special software algorithms.



What is computer vision?



Computer Vision System



How does computer vision work?

- ☐ **Machines interpret images as a series of pixels, each with their own set of color values.**
- ☐ **Each pixel's brightness in the image is represented by a single 8-bit number, ranging from 0 (black) to 255 (white).**
- ☐ **These numbers are what software sees when you input an image.**
- ☐ **This data is provided as an input to the computer vision algorithm that will be responsible for further analysis and decision making.**

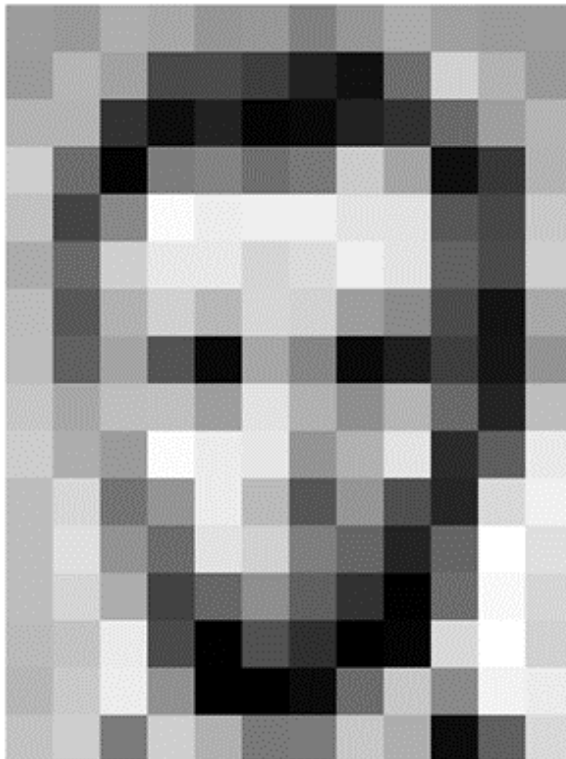
How does computer vision work?



0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

How does computer vision work?

Color values of individual pixels are converted into a simple array of numbers used as input for a computer vision algorithm.



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

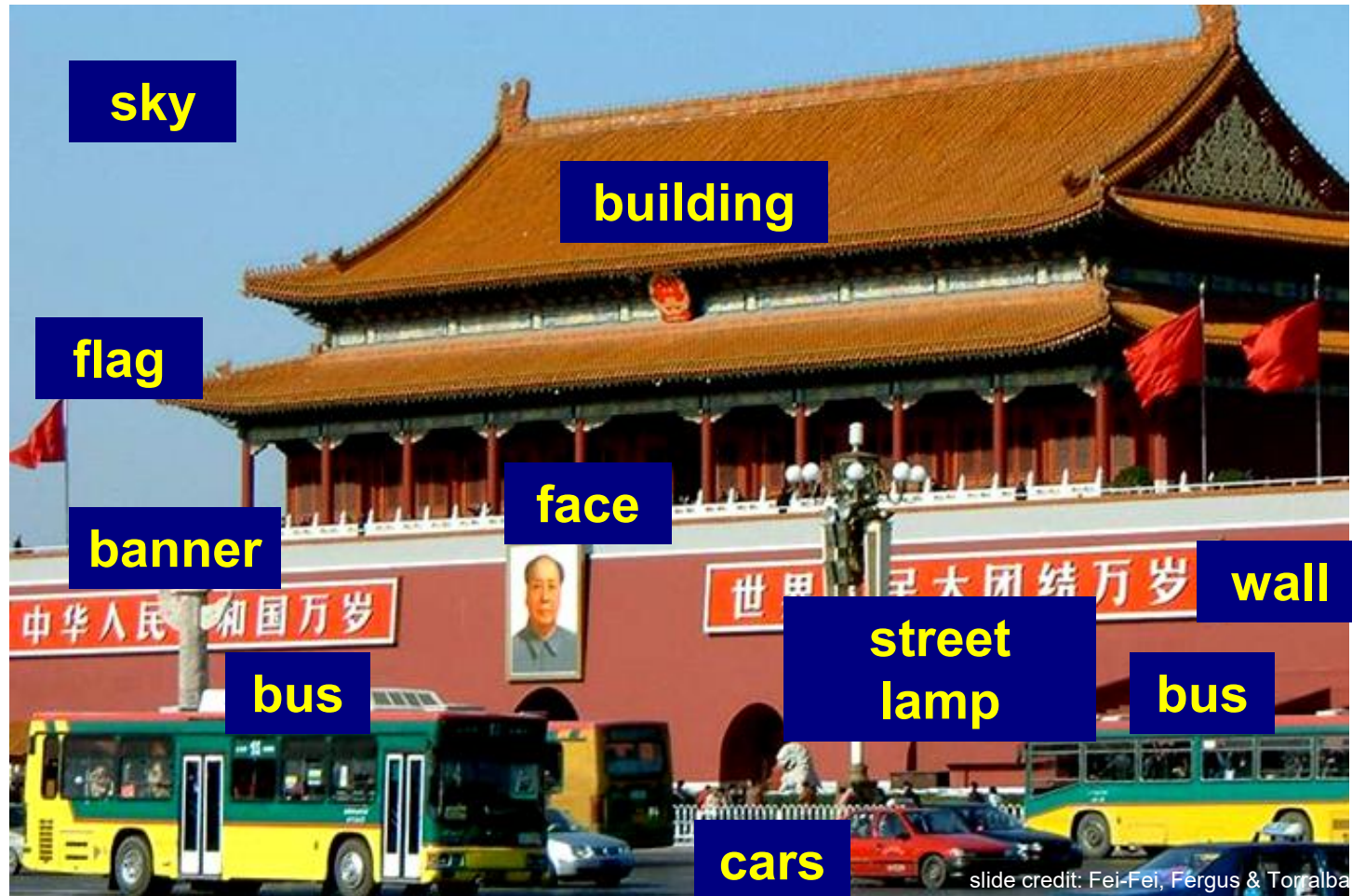
157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

The goal of computer vision

Recognize objects and people



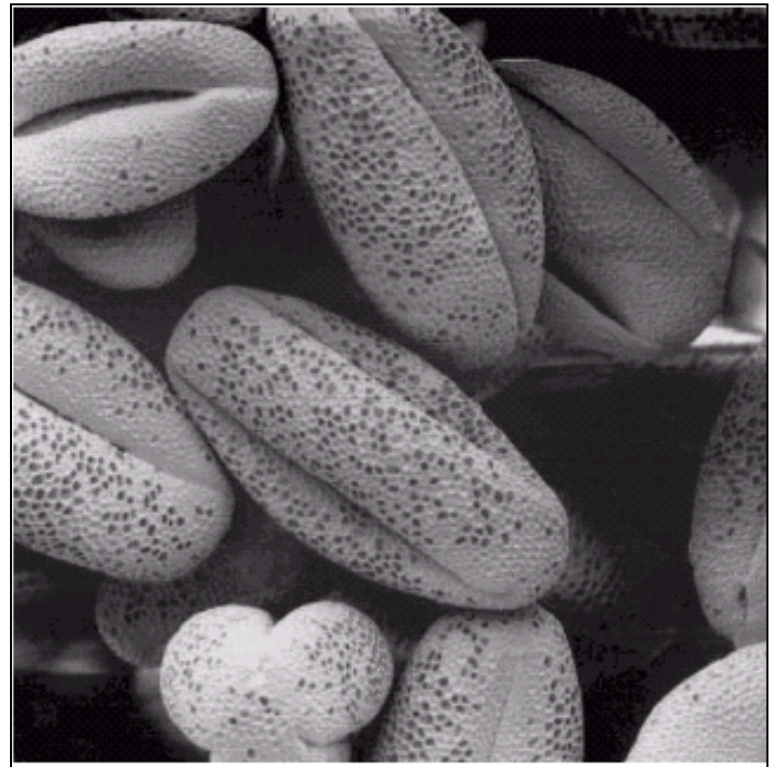
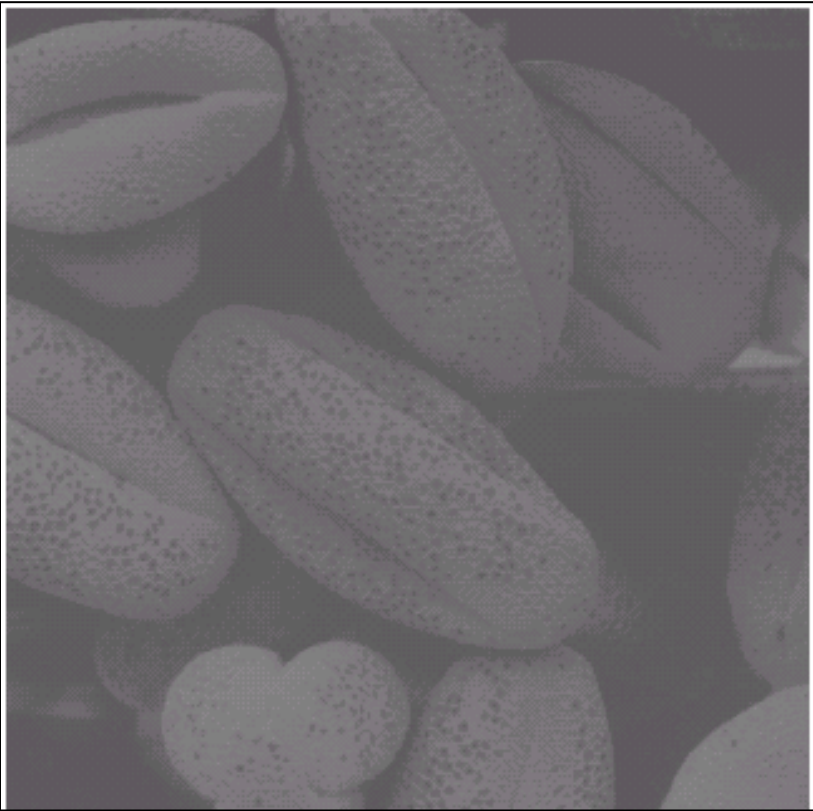
The goal of computer vision



Recognize objects and people

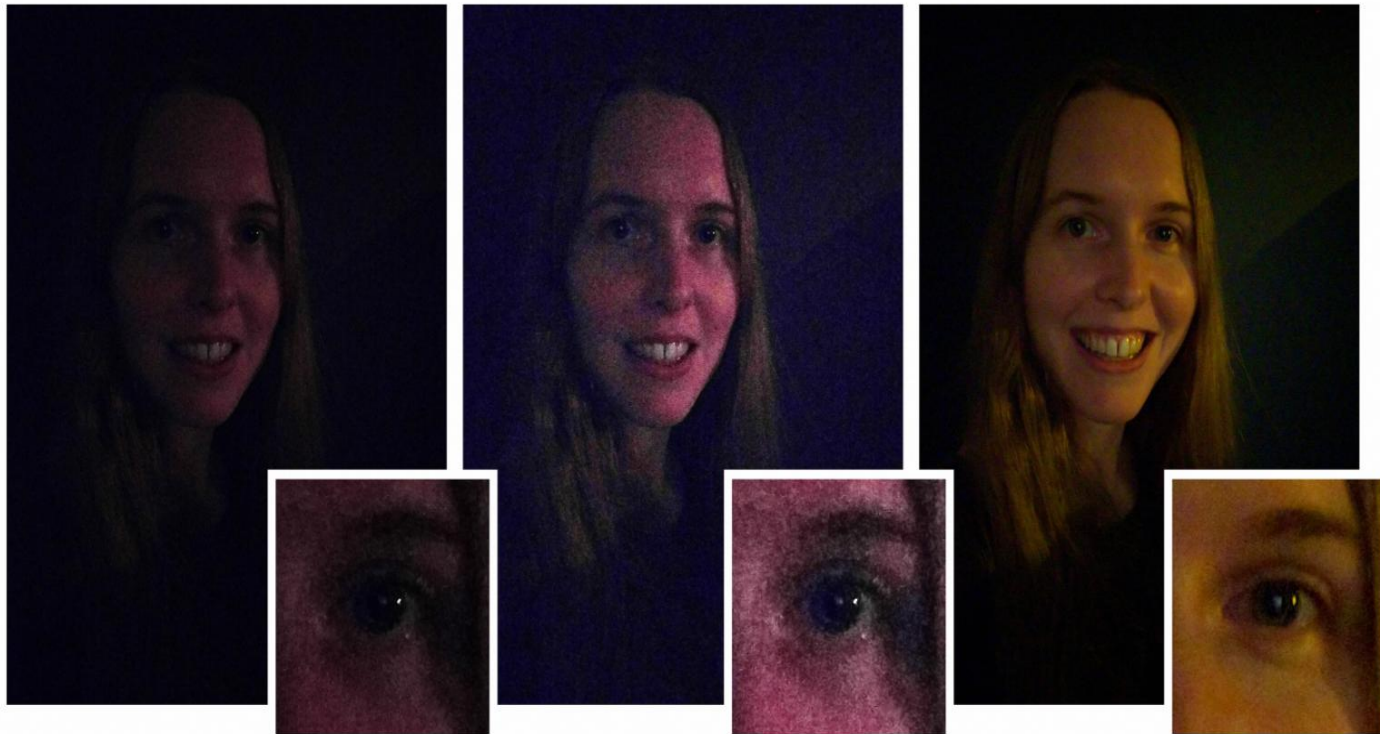
The goal of computer vision

“Enhance” images



The goal of computer vision

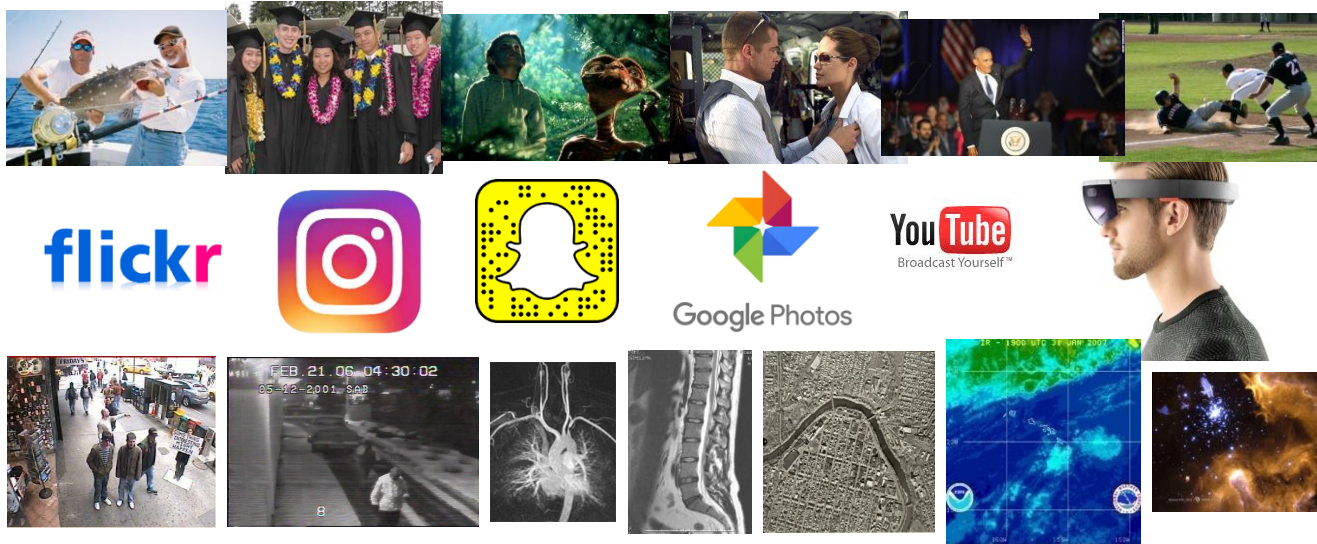
Improve photos (“Computational Photography”)



Low-light photography

Applications of Computer Vision

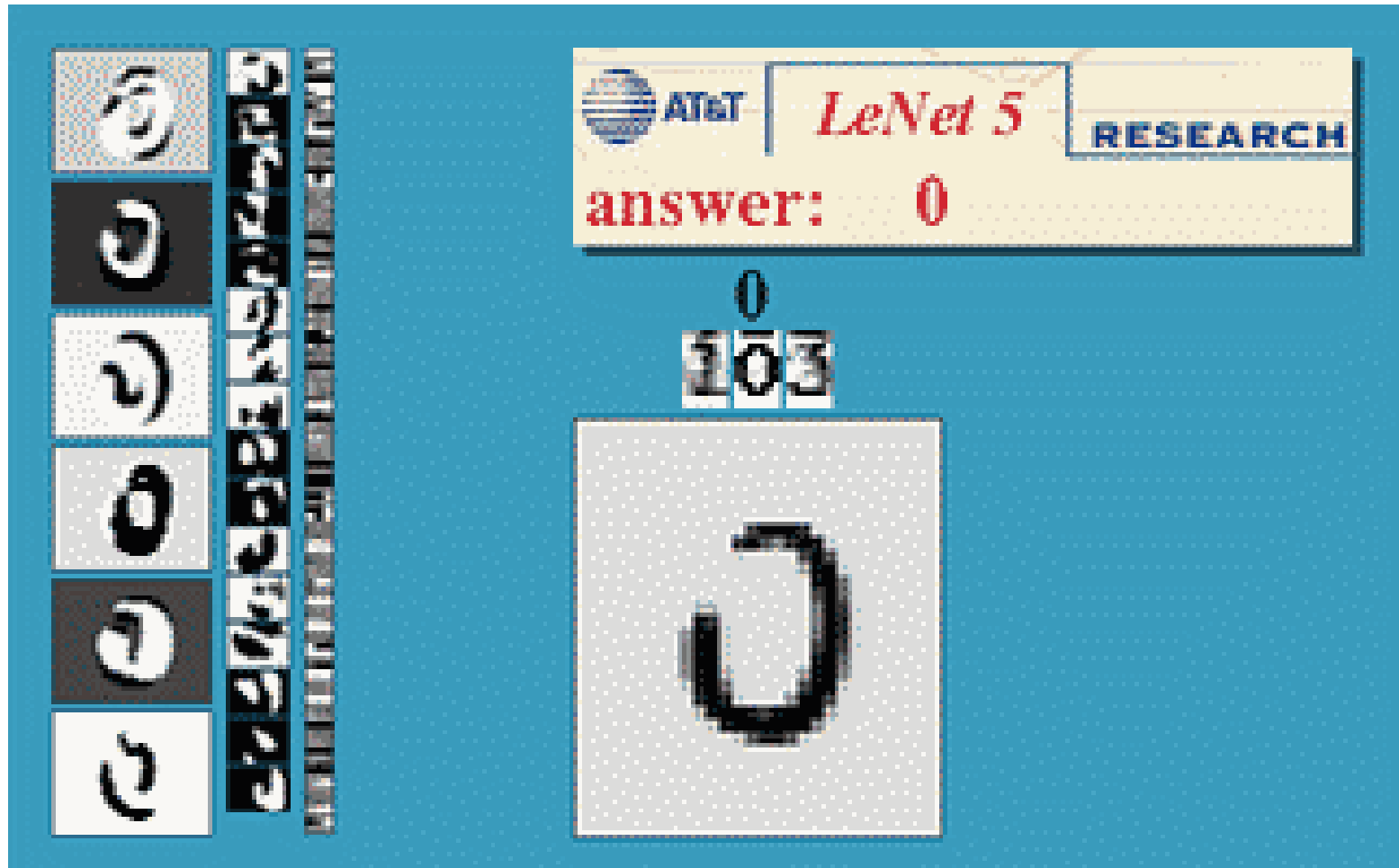
- Billions of images/videos captured per day



- Huge number of useful applications
- The next slides show the current state of the art

Applications of Computer Vision

Optical character recognition (OCR)



Applications of Computer Vision

Optical character recognition (OCR)



The screenshot shows a 'Check Entry' window with a check image and a data entry form. The check is from 'DataTrade Demo' for \$653.10, dated Feb 18, 2006. The form below the check contains fields for Batch No (1001), Sequence No (2), Routing No (111907445), Check Account No (72344999), Check No (100), and Check Amount (653.10). The Merchant Remit logo is visible at the bottom left.

Batch No	Sequence No	Routing No	Check Account No	Check No	Check Amount
1001	2	111907445	72344999	100	653.10

**Automatic check
processing**

4YCH428

4YCH428

4YCH428

License plate readers

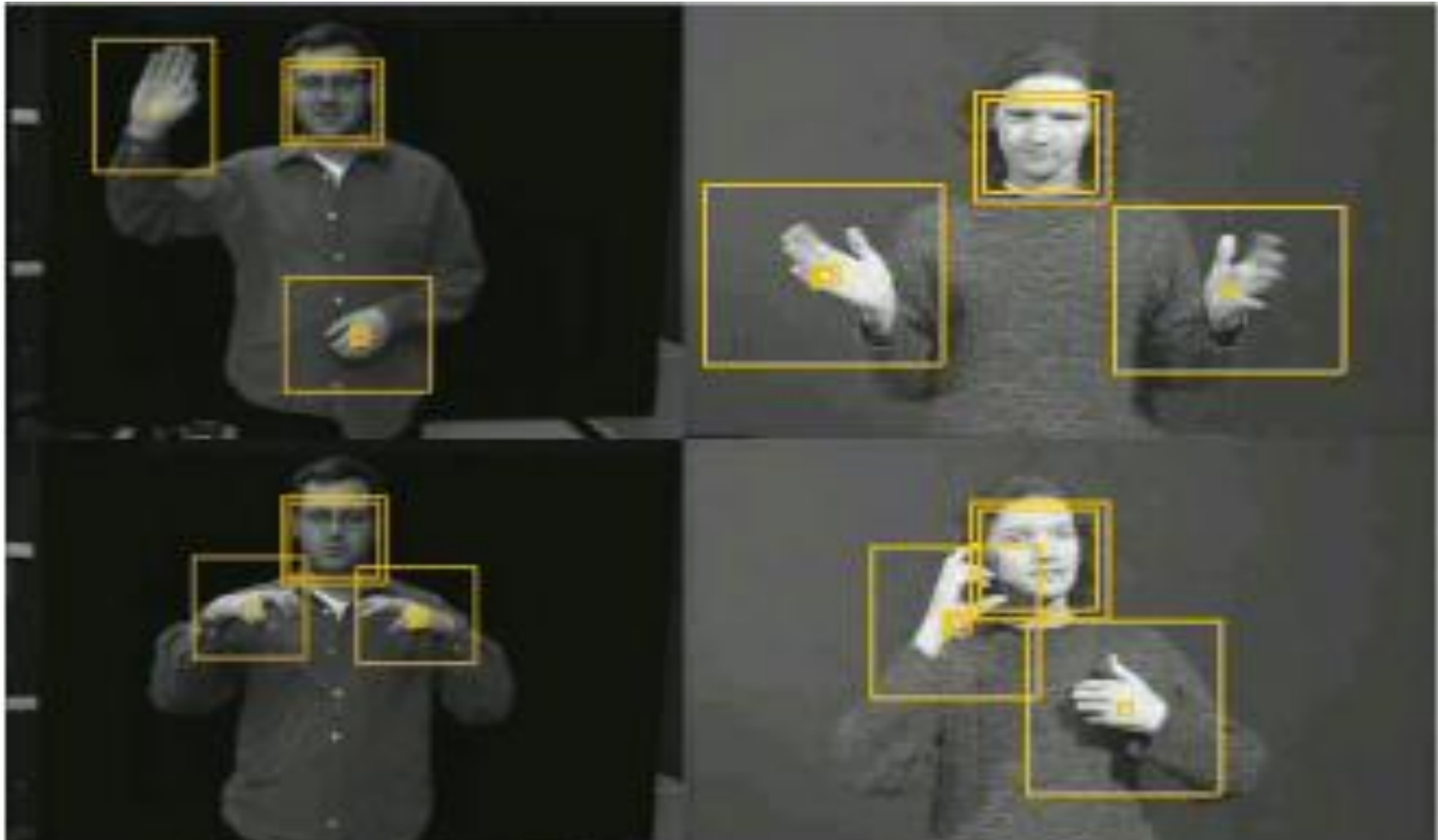
Applications of Computer Vision

Face detection



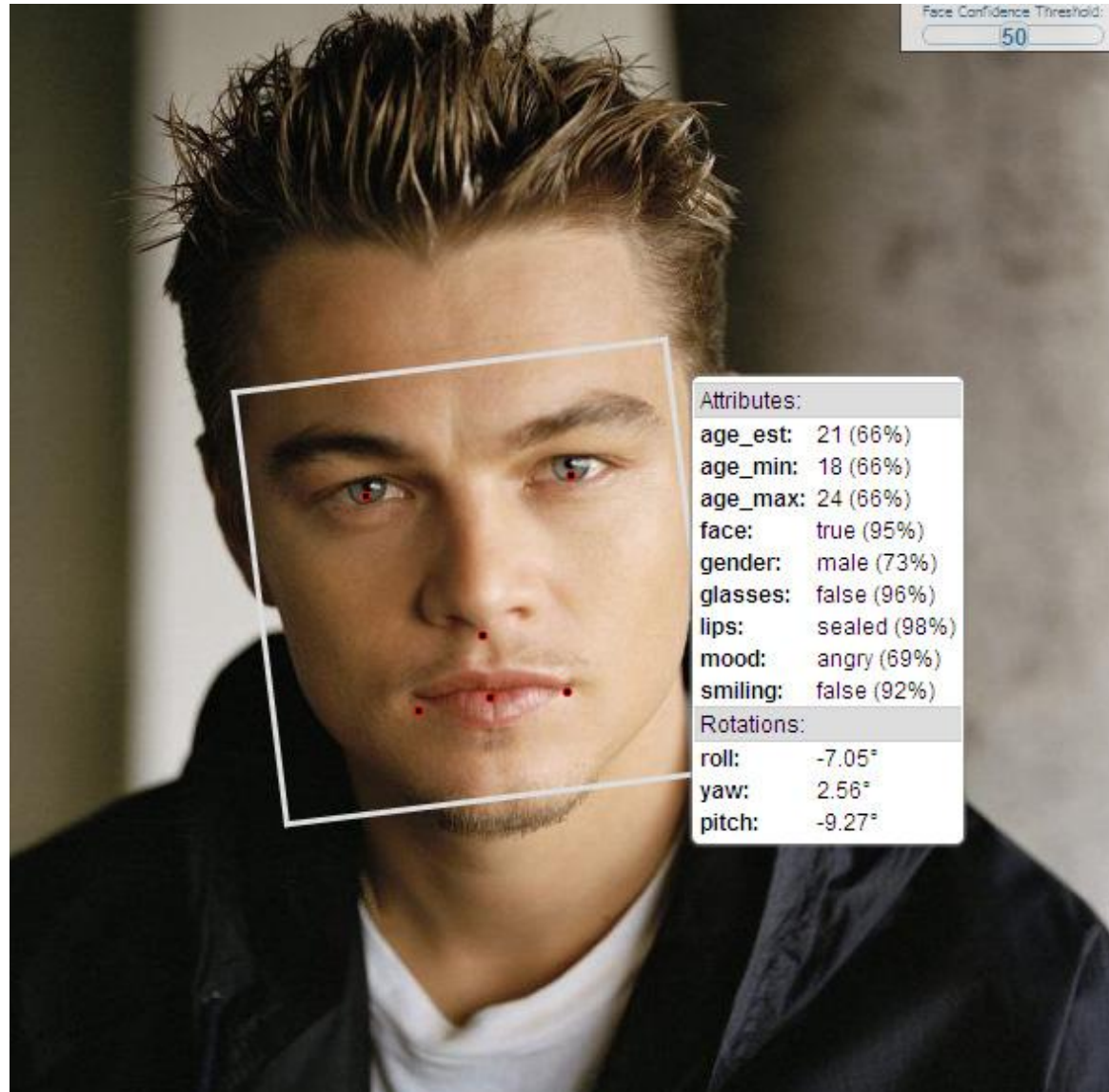
Applications of Computer Vision

Face recognition -Gesture recognition



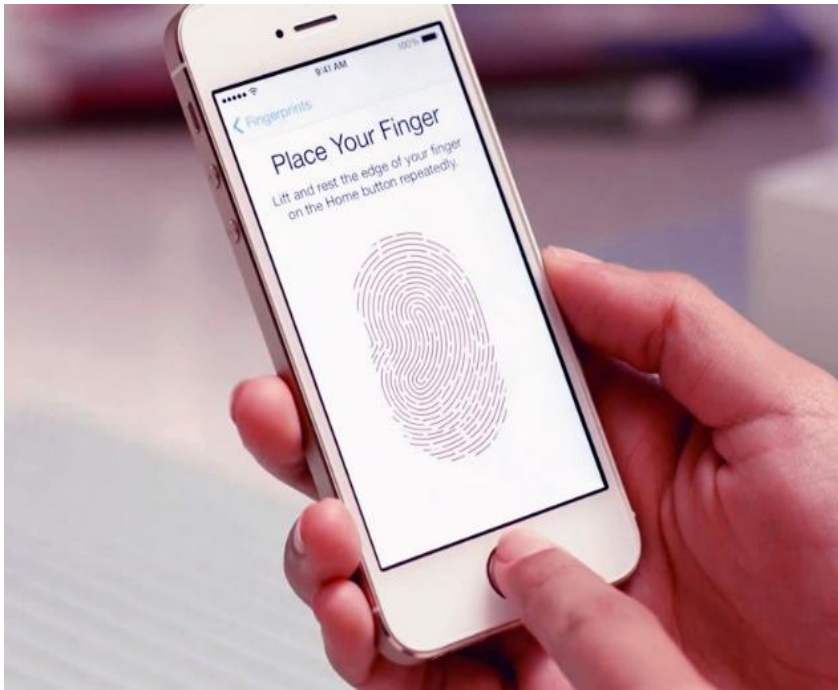
Applications of Computer Vision

Face analysis and recognition



Applications of Computer Vision

Login without a password



Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone

Applications of Computer Vision

Smart cars

The screenshot displays the Mobileye website with a top navigation bar containing 'manufacturer products' and 'consumer products'. The main banner features a car with three camera fields of view: 'rear looking camera', 'forward looking camera', and 'side looking camera', under the slogan 'Our Vision. Your Safety.'.

Below the banner are three product highlights:

- EyeQ Vision on a Chip**: Accompanied by an image of the EyeQ chip and a 'read more' link.
- Vision Applications**: Accompanied by an image of a pedestrian on a crosswalk and the text 'Road, Vehicle, Pedestrian Protection and more', with a 'read more' link.
- AWS Advance Warning System**: Accompanied by an image of a car icon on a circular display showing '0.8' and a 'read more' link.

On the right side, there are two sections:

- News**: Contains two headlines: 'Mobileye Advanced Technologies Power Volvo Cars World First Collision Warning With Auto Brake System' and 'Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end', with an 'all news' link.
- Events**: Contains two headlines: 'Mobileye at Equip Auto, Paris, France' and 'Mobileye at SEMA, Las Vegas, NV', with a 'read more' link.

- [Mobileye](#)
- Tesla Autopilot
- Safety features in many cars

▶▶ manufacturer products

consumer products ◀◀

Our Vision. Your Safety.

rear
looking
camera



forward
looking
camera

side looking camera

EyeQ Vision on a Chip



> read more

Vision Applications



Road, Vehicle,
Pedestrian Protection
and more

> read more

AWS Advance Warning System



> read more

- Safety features in many cars

> Mobileye Advanced Technologies
Volvo Cars Warns of Collision with Auto Brake System

> Volvo: New Collision Warning with Auto Brake Helps Prevent Accidents

> Mobileye at CES in Las Vegas, NV

> Mobileye at CES in Las Vegas, NV

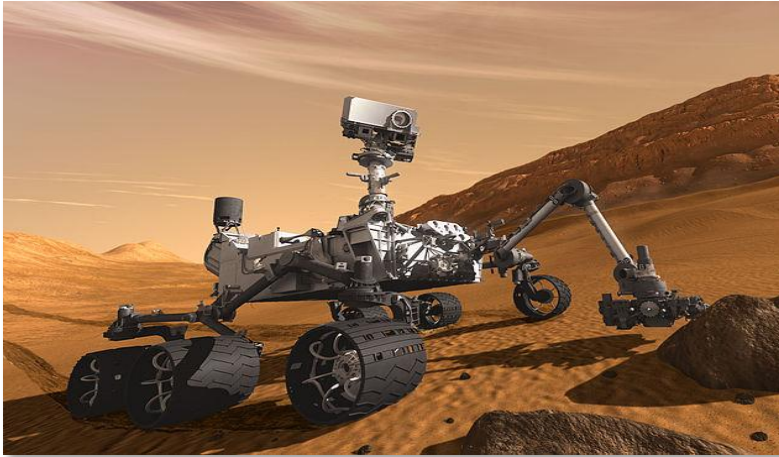
Applications of Computer Vision

Self-driving cars



Waymo

Applications of Computer Vision Robotics



NASA's Mars Curiosity Rover

↓



Amazon Picking Challenge

↓

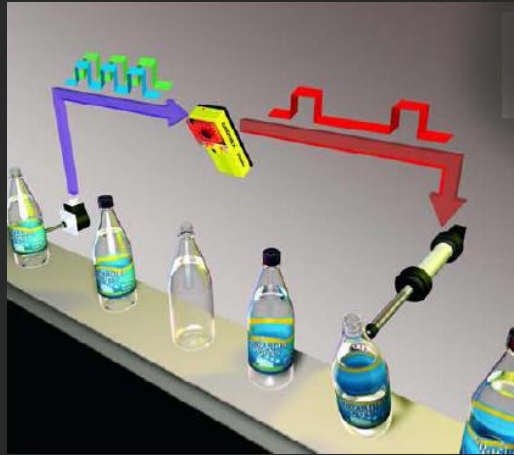


Amazon Prime Air



Amazon Scout

Applications of computer vision



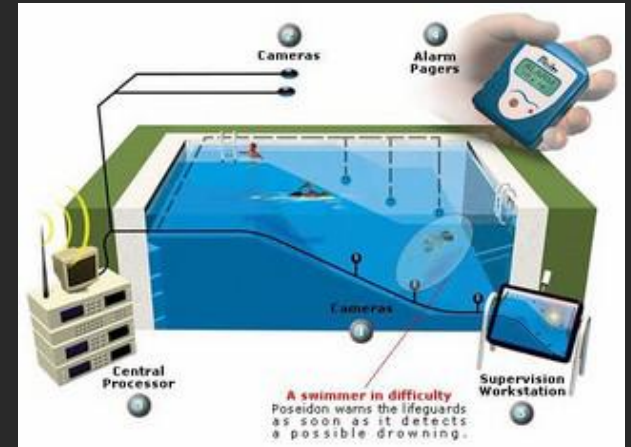
Factory inspection

الفحص بالمصنع



Reading license plates,
checks, ZIP codes

قراءة لوحات، الشيكات، رموز



Monitoring for safety

رصد للسلامة

Applications of computer vision



Surveillance مراقبة



Driver assistance
(collision warning, lane departure warning, rear object detection)



Autonomous driving, القيادة الذاتية
robot navigation ملاحة الروبوت

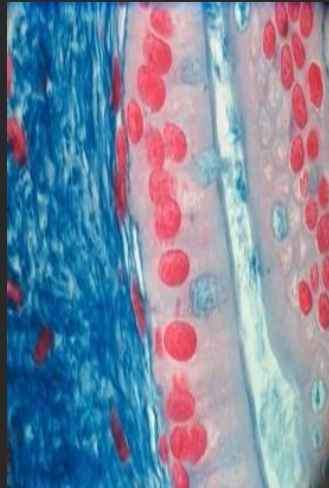
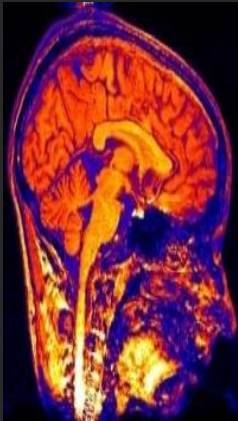
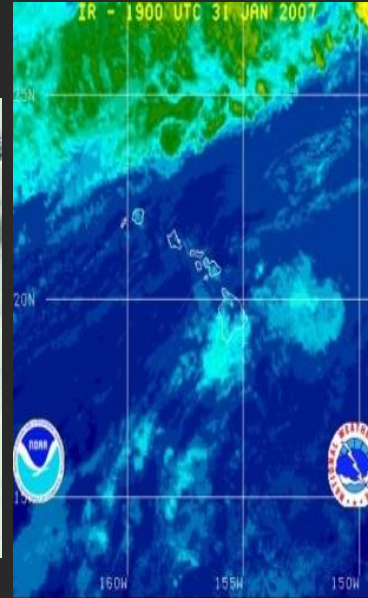
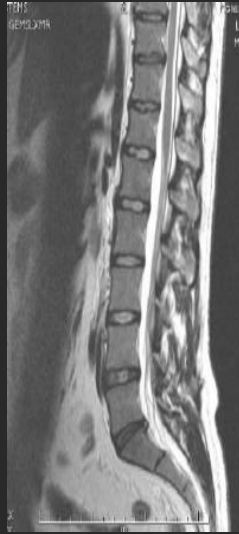
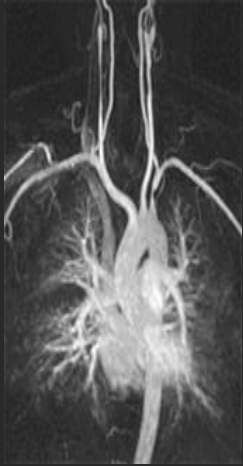
مساعدة السائق
(تحذير الاصطدام ، تحذير مغادرة حارة السير ،
وكشف أشياء من الخلف)

Surveillance and security



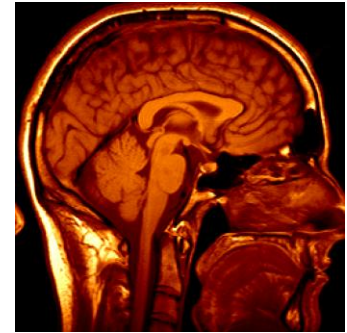
المراقبة والأمن

Medical and scientific images



الصور الطبية والعلمية

Medical imaging

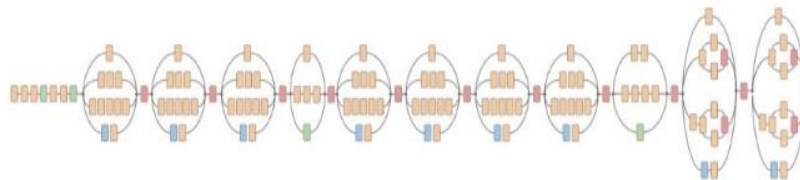


3D imaging
(MRI, CT)

Skin lesion image



Deep convolutional neural network (Inception v3)



- Convolution
- AvgPool
- MaxPool
- Concat
- Dropout
- Fully connected
- Softmax

Training classes (757)

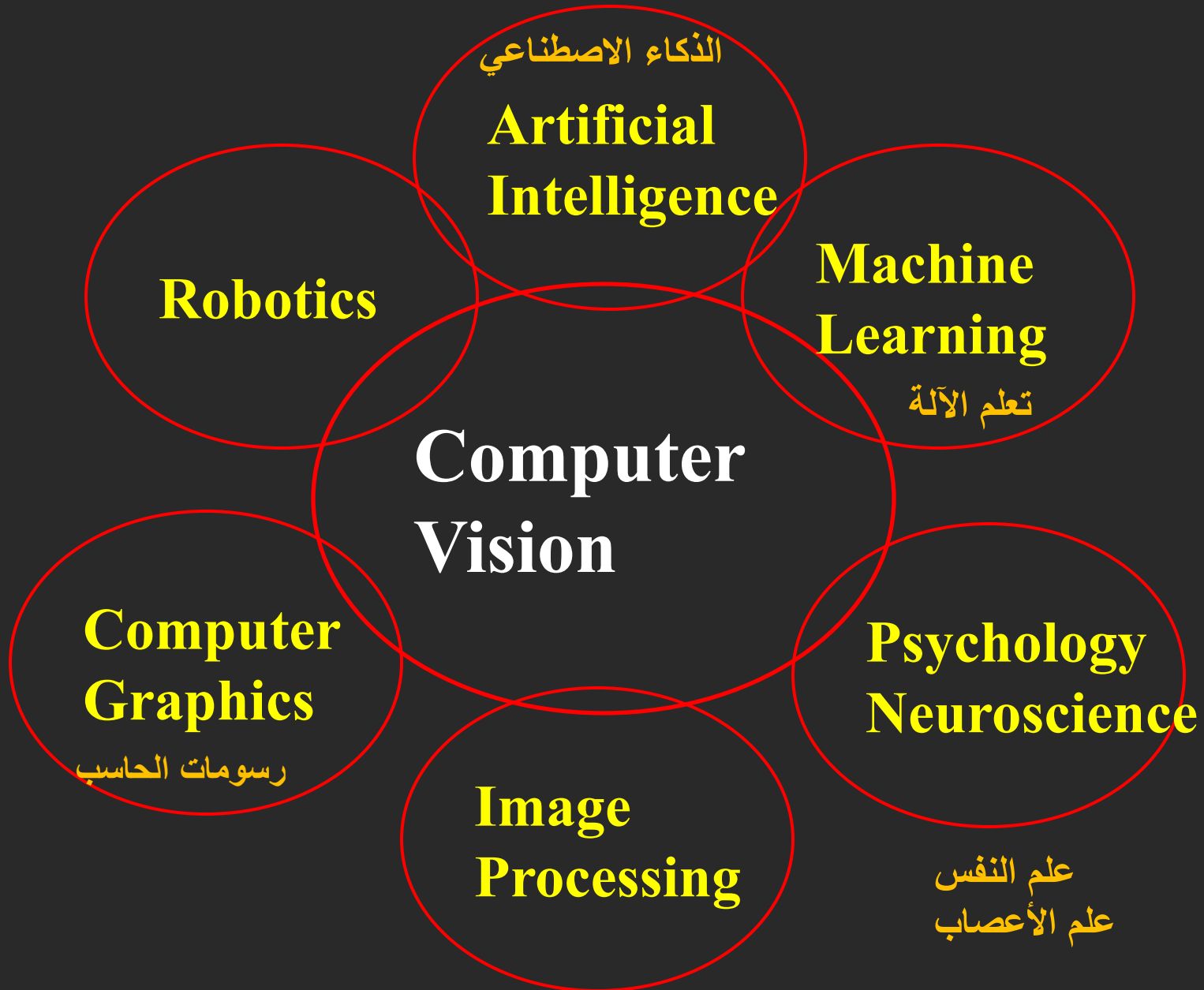
- Acral-lentiginous melanoma
- Amelanotic melanoma
- Lentigo melanoma
- ...
- Blue nevus
- Halo nevus
- Mongolian spot
- ...

Inference classes (varies by task)

- 92% malignant melanocytic lesion
- 8% benign melanocytic lesion

Skin cancer classification with deep learning <https://cs.stanford.edu/people/esteva/nature/>

Connections to other fields



Course overview

Course Aims

Course Contents

Study plan

Student Assessment

References

Course Aims

❑ *Upon successful completion of this course, students should be able to:*

- Understand the main characteristics of computer vision system.
- Analyze and interpret the content of images.
- Create and design a computer vision system for a real-world application.
- Implement and test computer vision algorithms using MATLAB programming.

Course Contents

- ❑ **Lecture** - The lectures will present the following topics: image acquisition, image filtering, image segmentation, feature extraction and object recognition.
- ❑ **Laboratory** - Students will implement and test computer vision algorithms introduced in lectures using **MATLAB programming**.
- ❑ **Project** - In the course project, students are required to design a real-world application of computer vision. Students will present their projects at the end of the class in a live-demo session.

Study plan

Week No.	Lecture			Tutorial/ Practical	
	Topic	Textbook (chapters)	No. Of Hours	Topic	No. Of Hours
1	Machine vision systems	Ch.1	2	Basic use of Matlab	2
2	Image acquisition: Image Sensors	Ch.2	2	Images and Matlab	2
3	Image acquisition: Digital Images	Ch.2	2	Image Display in Matlab	2
4	Image preprocessing: Filtering the image.	Ch.3	2	Filtering in Matlab	2
5	Image preprocessing: Histogram Equalization	Ch.3	2	Histograms in Matlab	2
6	Segmentation: Edge detection	Ch.4	2	Edges in Matlab	2
8	Mid - Term Exam				
7	Mid - Term Exam				

Study plan

Week No.	Lecture			Tutorial/ Practical	
	Topic	Textbook (chapters)	No. Of Hours	Topic	No. Of Hours
8	Segmentation: Global Segmentation	Ch.4	2	Edges in Matlab	2
9	Feature extraction: Boundary description	Ch.5	2	Edges in Matlab	2
10	Feature extraction: Fourier Descriptors.	Ch.6	2	Thresholding in Matlab	2
11	Feature extraction: Region Descriptors.	Ch.6	2	Thresholding in Matlab	2
12	Pattern classification: Neural Network approach.	Ch.7	2	Neural Network in Matlab.	2
13	Multilayer Feed-forward Neural Network	Ch.7	2	Practical Exam	

8 Mid - Term Exam

Final- Term Exam

14

Student Assessment

Methods	Weight	Week
Mid- Term Exam.	20	7
Quizzes and reports	10	All weeks
Practical Exam. & Project	10	13
Final-Term Exam	60	14
Total	100	

References

Course Notes:

Lectures in Computer Vision, prepared by Dr. Mahmoud Abu_Alfutuh.

Textbooks:

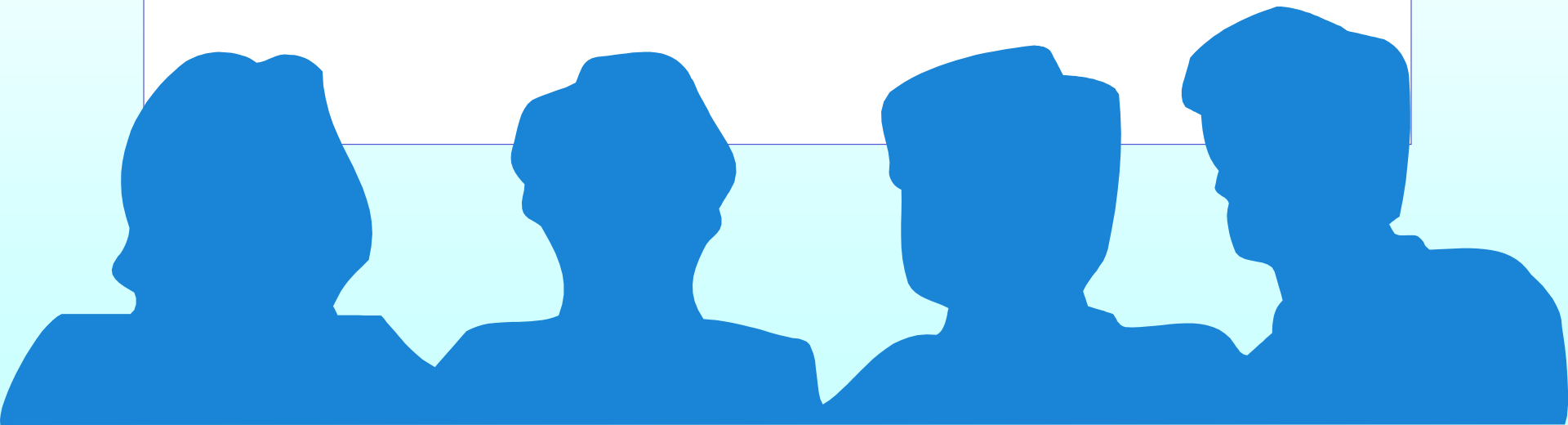
[1] Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd ed. © 2021.

[2] R. C. Gonzalez , R. Woods, S. Eddins , Digital image processing Using MATLAB, 3rd edition, Prentice-Hall, Inc Publishing company, 2020.

[3] G. J. Awcock and R. Thomas, “Applied Image Processing”, Macmilan press ltd London 1995.

Chapter 1

Machine Vision Systems



Introduction

- The goal of any vision system is to identify and locate a specific object in the scene.
- Machine vision system include data collection, preprocessing., feature extraction and matching.

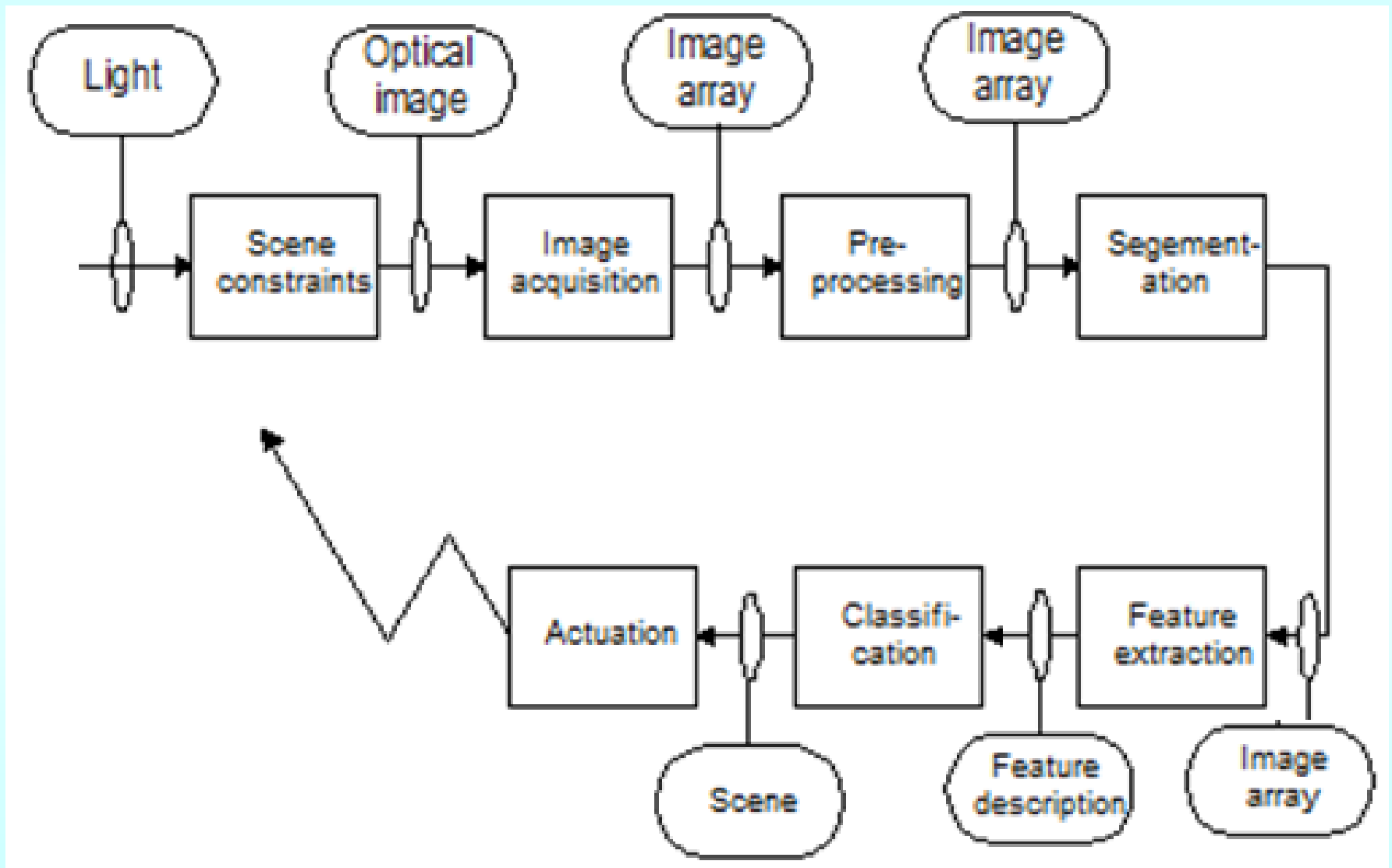
Generic Machine Vision System Model

- **Typical machine vision system identifies seven subsystems as follows:**

Machine Vision Sub-Systems

- **1. Scene Constraints.**
- **2. Image Acquisition.**
- **3. Preprocessing.**
- **4. Segmentation.**
- **5. Feature Extraction.**
- **6. Classification.**
- **7. Actuation**

Generic Machine Vision System Model



1.Scene Constraints القيود

Constraints might include:

- The number of objects.
- Installation of cameras
- Special lighting conditions

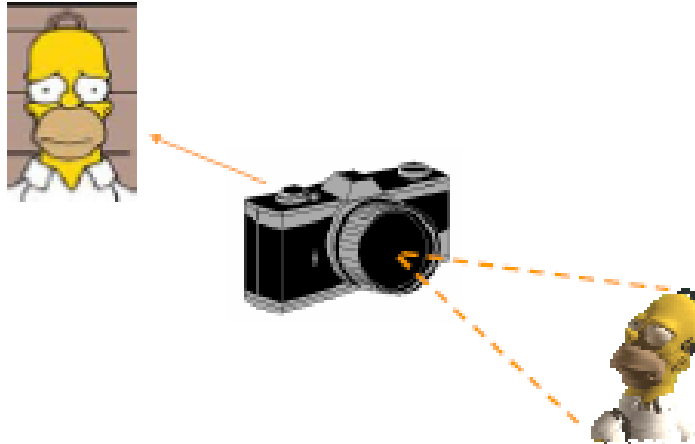
1. Scene Constraints

Achieving good control over the scene constraints will result in:

- **a less complex,**
 - **more reliable,**
 - **and less expensive**
- vision system implementation.**

2. Image Acquisition الحصول على الصورة

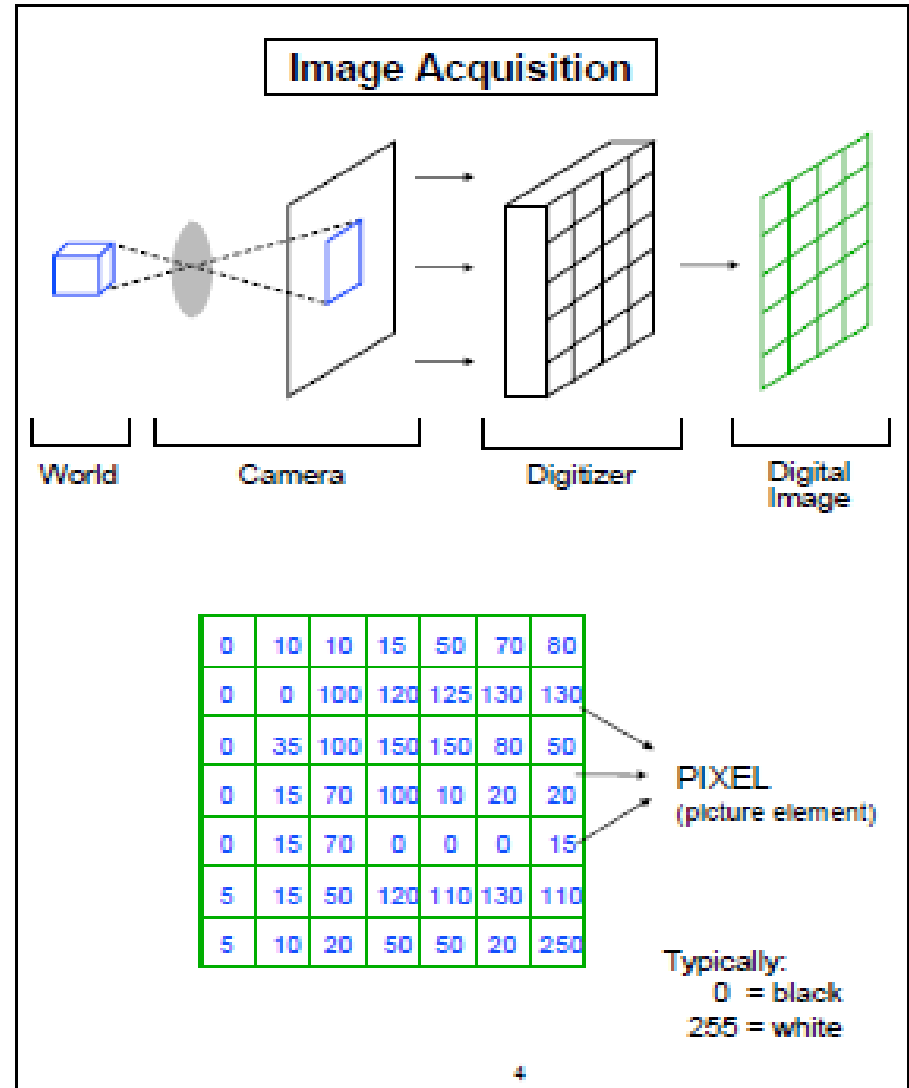
- Cameras are devices used to transform optical images into electrical signals.



- An image can be defined as a 2 variable function $I(x,y)$.
- For each position (x,y) in the projection plane, $I(x,y)$ defines the light intensity at this point.

2. Image Acquisition

A digital image is obtained by quantizing the output signals obtained from image acquisition devices.



2. Image Acquisition

Three types of images:

- Binary images

$$I(x,y) \in \{0, 1\}$$

- Gray-scale images

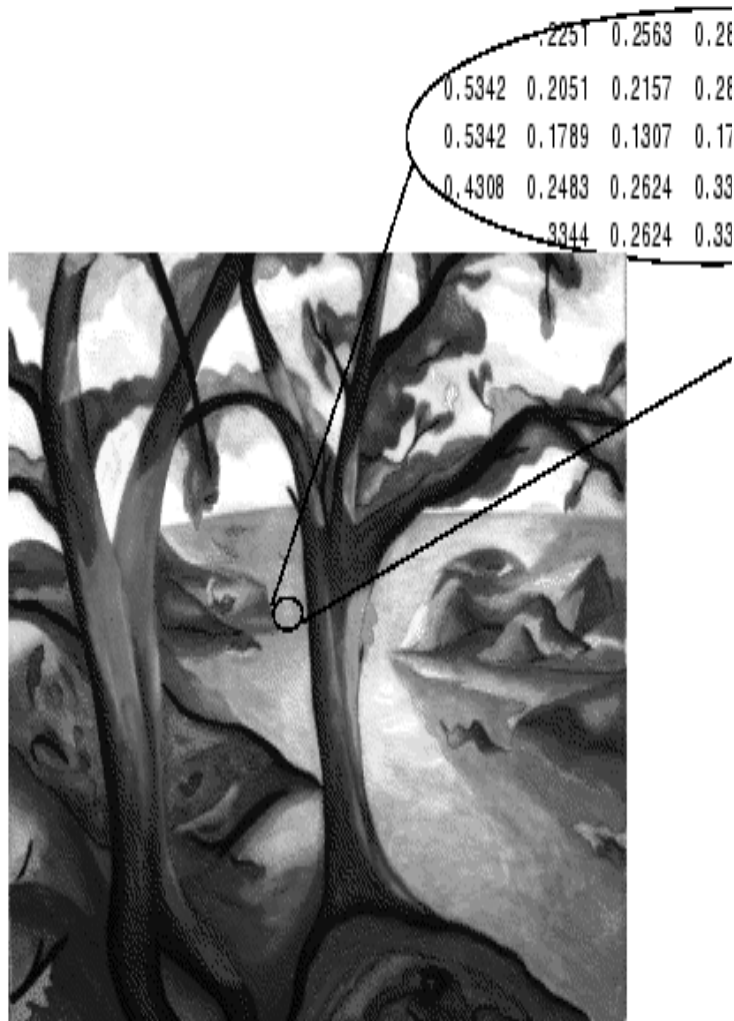
$$I(x,y) \in [a, b]$$

- Color Images

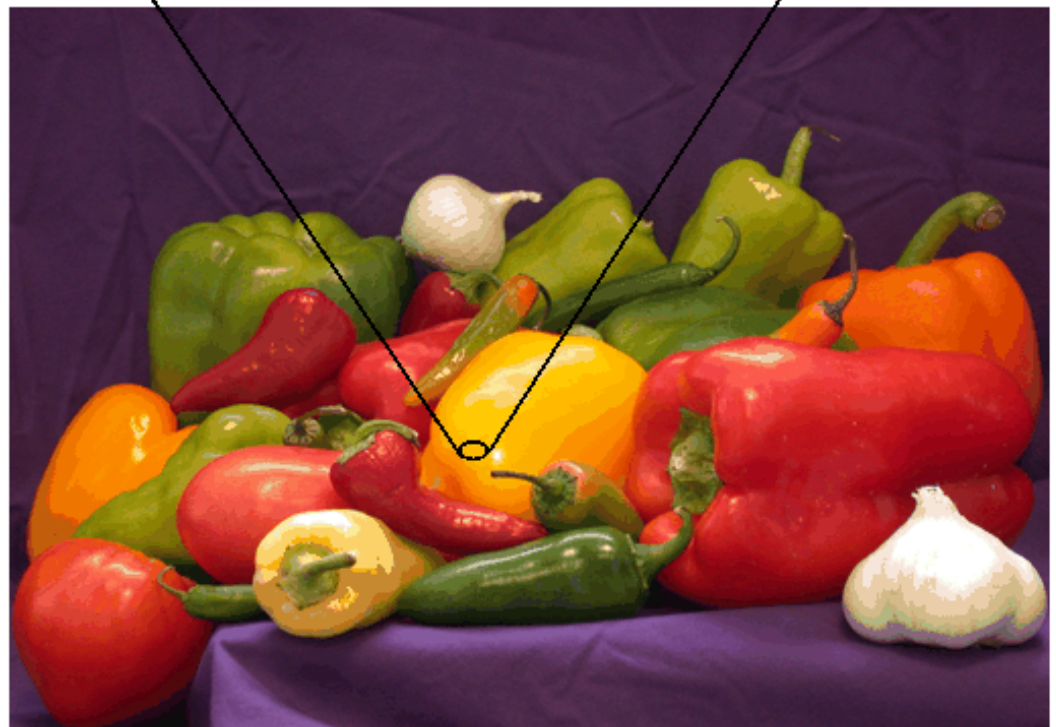
$$IR(x,y) \quad IG(x,y) \quad IB(x,y)$$



2. Image Acquisition



0.2235	0.1294	Blue	0.4196	0.4196	0.4196	0.4196
0.5804	0.2902	0.0627	0.2902	0.2902	0.4824	0.4824
0.5804	0.0627	0.0627	0.0627	0.2235	0.2588	0.2588
0.5176	0.1922	0.0627	Green	0.1922	0.2588	0.2588
0.5176	0.1294	0.1608	0.1294	0.1294	0.2588	0.2588
0.5176	0.1608	0.0627	0.1608	0.1922	0.2588	0.2588
0.5490	0.2235	0.5490	Red	0.7412	0.7765	0.7765
0.5490	0.3882	0.5176	0.5804	0.5804	0.7765	0.7765
0.490	0.2588	0.2902	0.2588	0.2235	0.4824	0.2235
0.2235	0.1608	0.2588	0.2588	0.1608	0.2588	0.2588
0.2588	0.1608	0.2588	0.2588	0.2588	0.2588	0.2588



3. Preprocessing

Preprocessing includes operations to:

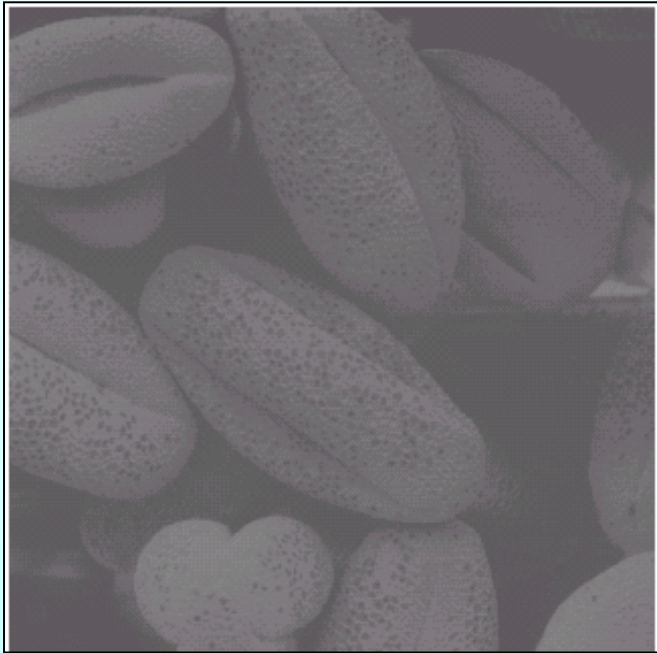
- 1-Clean up noisy images,



3. Preprocessing

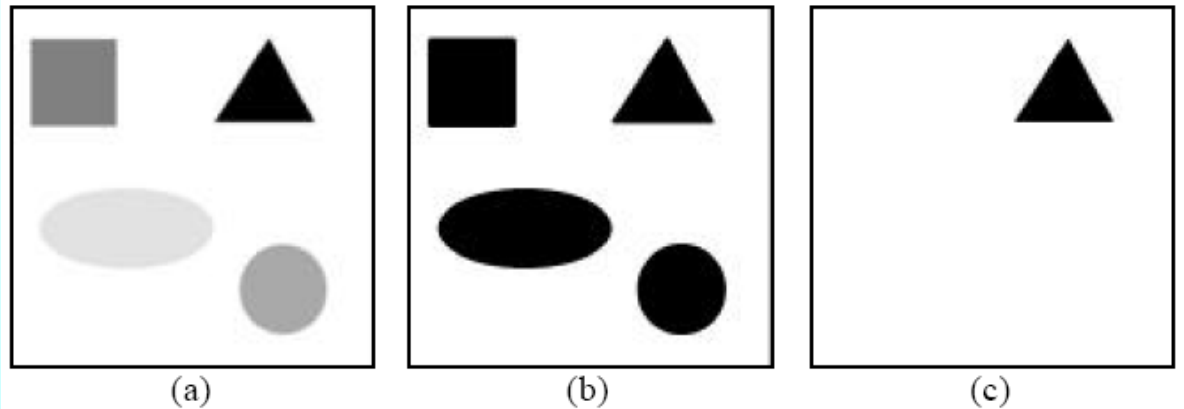
Preprocessing includes operations to:

- 2-Highlight particular features of interest.



4. Segmentation

- **Segmentation** refers to the process of partitioning an image into regions or segments.
- Image segmentation is used to locate objects and boundaries in images.

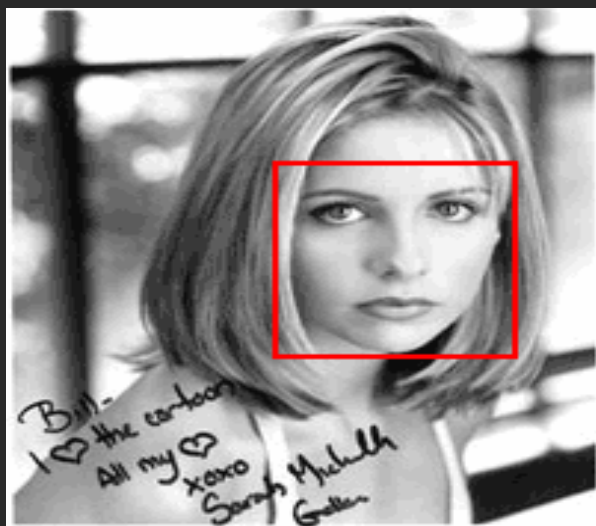


5. Feature Extraction

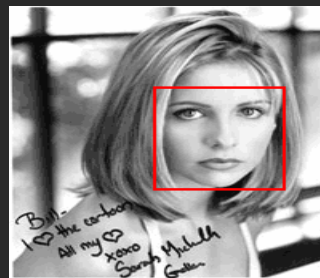
- Features are used to describe the object prior to the task of classification.
- Feature extraction operates on two-dimensional image arrays but produces a list of descriptions, or a 'feature vector'.
- Features are gleaned from either:
 - The boundary properties of the shape.
 - Or the internal properties of the shape.

5. Feature Extraction

The features should be invariant to position, orientation and scale of the object.



Face detection



Face detection



Face detection

6. Classification

- **Classification is the process of correctly assigning unknown pattern to their respective pattern class.**
- **This process utilizes the object features.**

6. Classification

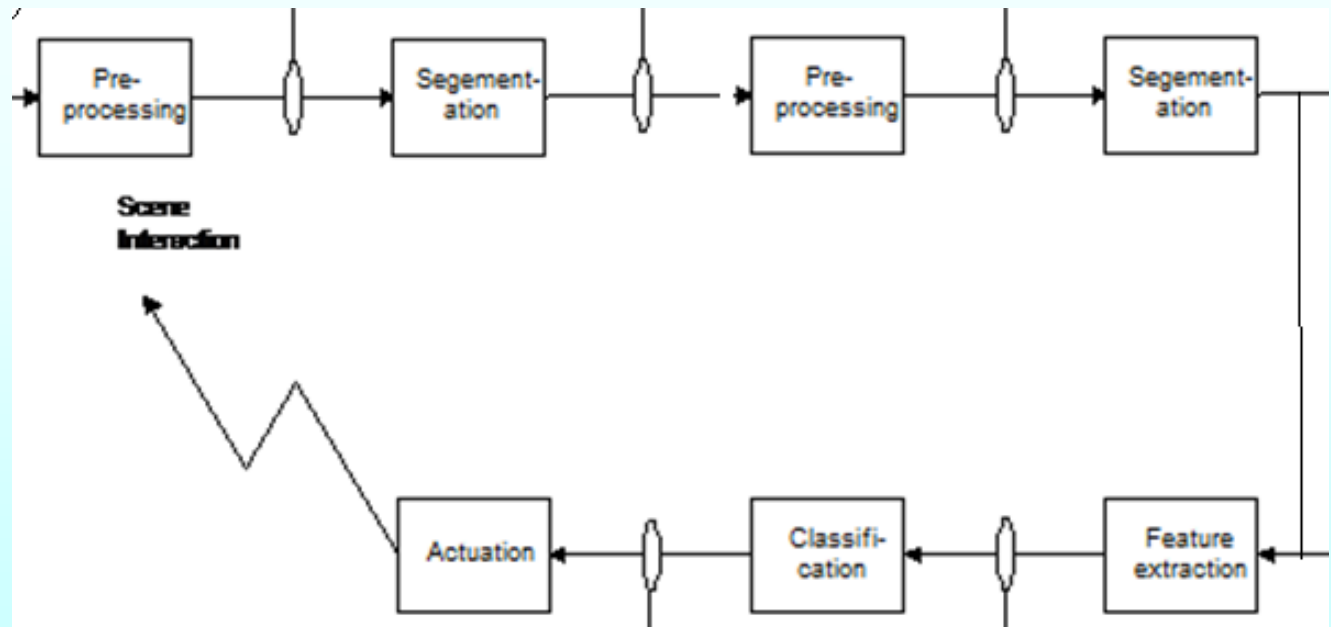
Three basic approaches to pattern classification:

- 1. Statistical pattern classification.**
- 2. Syntactic pattern classification.**
- 3. Artificial neural networks.**

7. Actuation

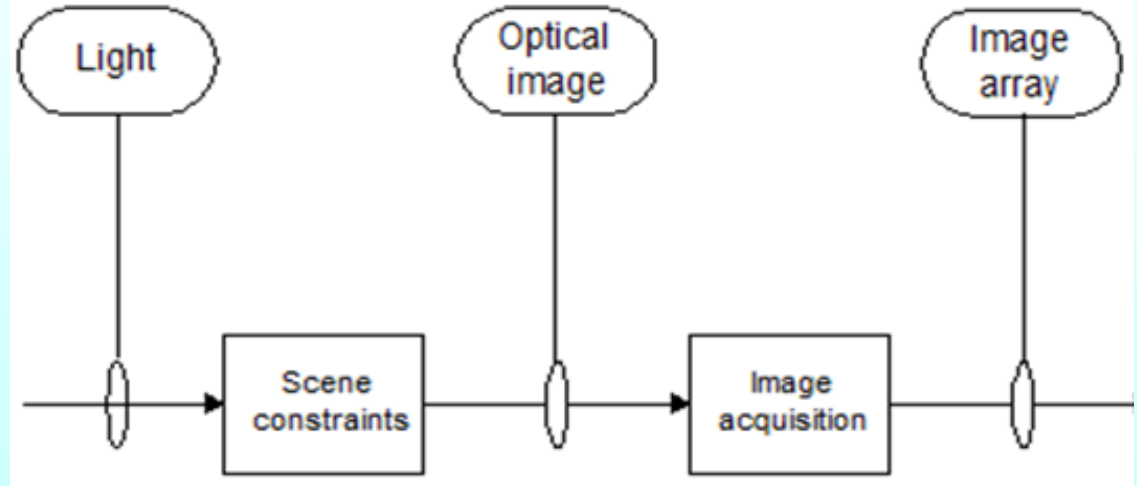
- The actuation sub-system provides a means of closing the loop

Allowing interaction with the original scene.



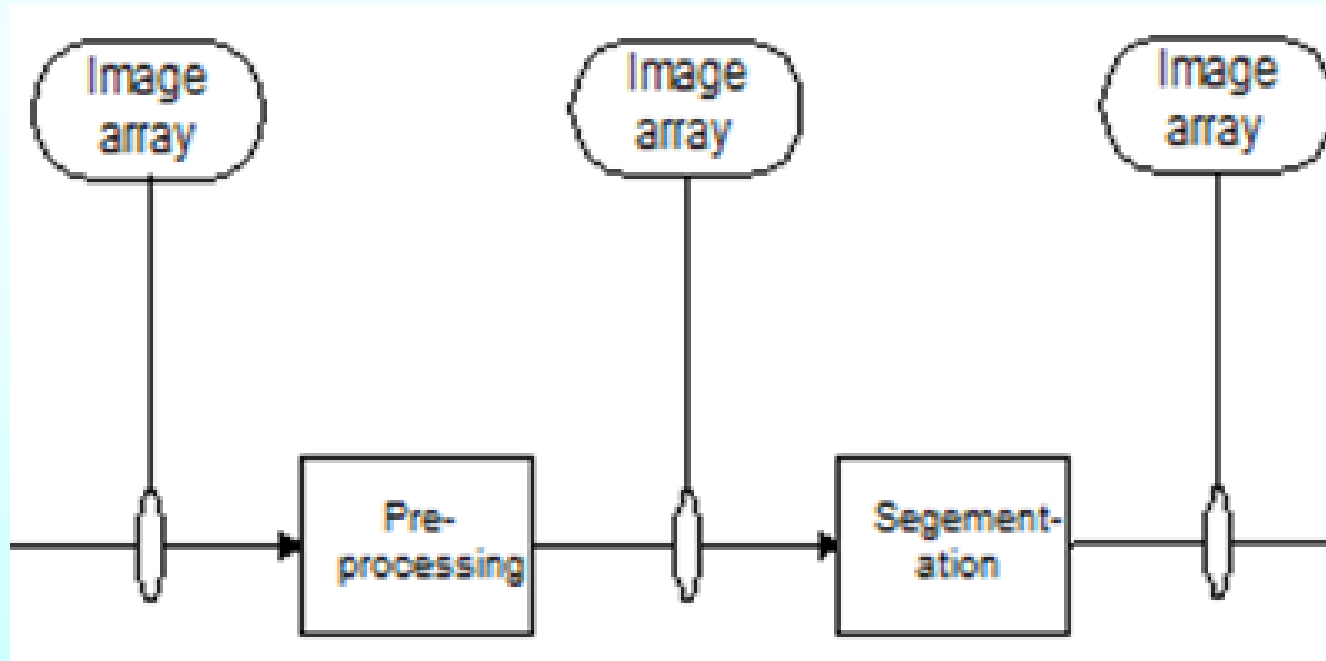
Information Format

- Information format available at each stage is as follows:
- After the image is acquired, the information is in the form of an array of sampled data (the image array).



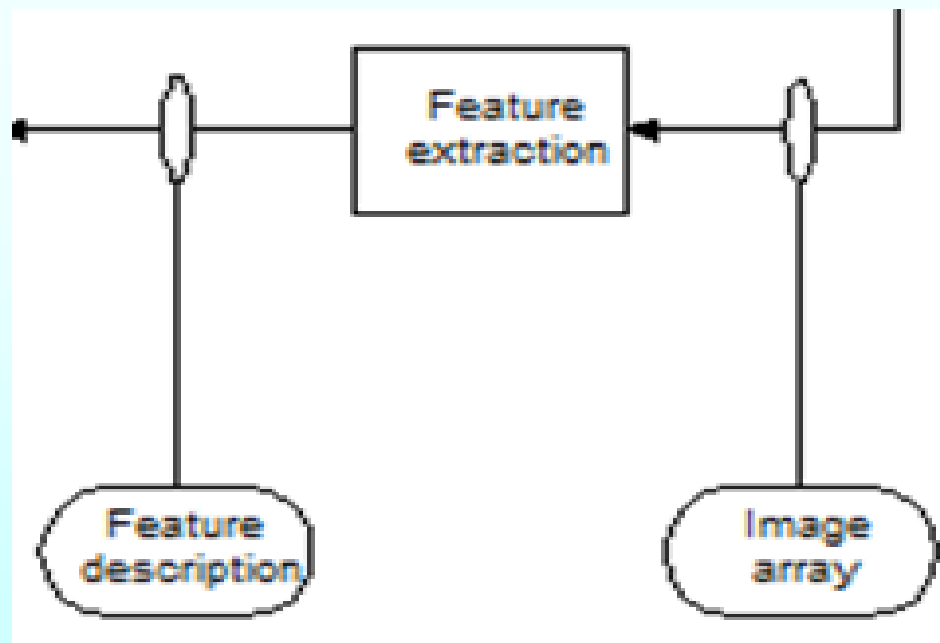
Information Format

- The processing and segmentation operations upon this array produce the same image format (a modified image array).



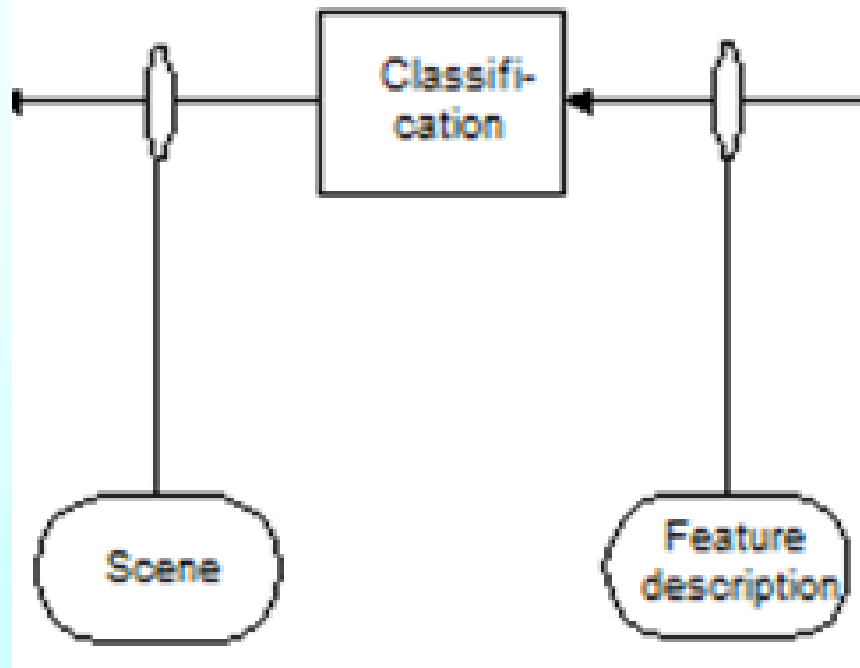
Information Format

- The feature extraction process uses the image array data in order to produce a list of feature descriptions.



Information Format

- The classification operation produce some precise descriptions of the scene or objects within it.



Thank you for listening